

Minh Nhat Tran

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Summary

First-year Ph.D. student in Geophysics at the University of Houston, specializing in seismic wave propagation, full-waveform inversion, and DAS, with a focus on computational and machine-learning methods.

Research Interests

- Full-waveform inversion (FWI).
- Distributed Acoustic Sensing (DAS) and field-data FWI.
- Implicit neural representations and machine learning for FWI and wave propagation.

Education

University of Houston <i>Ph.D. student in Geophysics</i>	Houston, Texas, USA <i>January 2026 – Present</i>
University of Florida <i>Ph.D. student in Geosystems Engineering</i>	Gainesville, Florida, USA <i>May 2024 – January 2026</i>
Ho Chi Minh City University of Technology – VNU-HCM <i>B.E., Civil Engineering</i>	Ho Chi Minh City, Vietnam <i>August 2018 – November 2022</i>

Research Experience

University of Houston <i>Research Assistant</i>	Houston, Texas, USA <i>January 2026 – Present</i>
• Developed a velocity–strain finite-difference solver in JAX for Distributed Acoustic Sensing (DAS) data. • Applied implicit FWI to field seismic data.	
University of Florida <i>Research Assistant</i>	Gainesville, Florida, USA <i>May 2024 – January 2026</i>
• Developed 2D elastic FWI codes with graphical user interfaces (GUIs): SH–Love for site characterization and PSV/Rayleigh for land-streamer data, with MASW inversion for initial models. • Developed a source-encoding approach to accelerate 3D elastic FWI.	
Ho Chi Minh City University of Technology – VNU-HCM <i>Research Assistant</i>	Ho Chi Minh City, Vietnam <i>August 2022 – May 2024</i>
• Modeled seismic soil–structure interaction using FEA with the Domain Reduction Method (DRM). • Applied machine learning (ANN, XGBoost) for parametric studies and predictive modeling.	

Publications

- [1] Nguyen, K. T., & **Tran, M. N.** (2026). Seismic responses of soil–twin tunnels system subjected to Rayleigh wave propagation. *Soil Dynamics and Earthquake Engineering*, 208, 110316.
DOI: [10.1016/j.soildyn.2026.110316](https://doi.org/10.1016/j.soildyn.2026.110316)

- [2] Chen, R., Tran, K. T., McVay, M., Yang, K., & **Tran, M. N.** (2025). Characterization of in-situ rock density with SH and Love-wave tomography: field data application. *Journal of Geophysics and Engineering*, gxaf096. DOI: 10.1093/jge/gxaf096
- [3] Lai, V. Q., **Tran, M. N.**, Shiau, J., Keawsawasvong, S., Le, B. V., Nguyen, T. K., & Le, D. Q. (2023). Uplift resistance of caisson foundation in NGI-ADP soil using FEA and ANN. *Advances in Engineering Software*, 184, 103513. DOI: 10.1016/j.advengsoft.2023.103513
- [4] Lai, V. Q., Nguyen, T. K., Shiau, J., Keawsawasvong, S., Bui, T. S., & **Tran, M. N.** (2023). Coupling FEA with XGBoost model for estimating uplift resistance of circular anchor in NGI-ADP soils. *Geotechnical and Geological Engineering*, 1–15. DOI: 10.1007/s10706-023-02556-9
- [5] Tran, D. T., **Tran, M. N.**, Lai, V. Q., et al. (2023). Advanced FELA-ANN framework for developing 3D failure envelopes for strip foundations on anisotropic clays. *Modeling Earth Systems and Environment*. DOI: 10.1007/s40808-023-01906-z

Conference Presentations

- [6] **Tran, M. N.**, Li, J. (2026). Strain-based full waveform inversion with automatic differentiation for distributed acoustic sensing. (IMAGE 2026, Poster)
- [7] **Tran, M. N.**, Boonyatee, T., Le, A. T., Lai, V. Q. (2023). Bearing capacity of circular footing on $c-\phi$ slope. *Proceedings of the 5th International Conference on Geotechnics for Sustainable Infrastructure Development – GEOTEC HANOI 2023*. DOI: 10.1007/978-981-99-9722-0_193
- [8] **Tran, M. N.**, Keawsawasvong, S., Lam, S., Le, A. T., Luong, V. H., Lai, V. Q. (2024). Predicting bearing capacity of strip footing on slope using a soft computing technique. In: Reddy, J. N., Wang, C. M., Luong, V. H., Le, A. T. (eds), *Proceedings of the Third International Conference on Sustainable Civil Engineering and Architecture (ICSCEA 2023)*, Lecture Notes in Civil Engineering, vol. 442. Springer, Singapore. DOI: 10.1007/978-981-99-7434-4_127

Technical Reports

- [1] Tran, K. T., McVay, M., **Tran, M. N.**, Yang, B., & Yang, K. (2024). *Determination of in-situ rock density and strength with SH-Love wave tomography*. Final Report, Florida Department of Transportation.

Selected Projects

- **Implicit FWI** — Developed a differentiable FWI workflow combining SIREN-based velocity parameterization in JAX with a Devito adjoint-state solver. [GitHub](#)
- **DAS-FWI** — Developing elastic FWI directly using Distributed Acoustic Sensing (DAS) strain measurements.

Skills

- **Software:** SPECFEM2D.
- **Programming languages:** Python, MATLAB, C++.

Awards

- 2026 Outstanding Graduate Work in Geophysics Award

Reference

Dr. Jiaxuan Li, Ph.D. University of Houston | jxli@cougarnet.uh.edu